

TASK 05: E-Commerce Return Rate Investigation

1. Problem Statement

E-commerce businesses process thousands of customer orders every day. While online shopping offers convenience, product returns create significant operational and financial challenges. High return rates increase logistics costs, inventory handling expenses, and customer service workload. Understanding the factors that contribute to product returns is essential for improving customer satisfaction and reducing business losses.

The objective of this project is to perform statistical and exploratory data analysis on e-commerce order data to identify patterns and factors influencing product returns. By analyzing customer behavior, product categories, pricing, and payment methods, the study aims to provide actionable business recommendations for reducing return rates.

2. Dataset Description

The dataset contains information about customer purchases, product details, and return behavior in an e-commerce environment.

Dataset Features

- Customer ID – Unique identifier for each customer.
- Product Category – Category of the purchased product.
- Product Price – Price of the purchased product.
- Quantity – Number of units purchased.
- Total Purchase Amount – Total amount spent on the order.
- Payment Method – Method used for payment.
- Customer Age – Age of the customer.
- Returns – Indicates whether the product was returned (1 = Returned, 0 = Not Returned).
- Gender – Gender of the customer.
- Churn – Indicates whether the customer left the platform.

Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked dataset dimensions and column information.
- Identified and handled missing values.
- Removed duplicate records.

Key Metrics

- Total Orders
- Total Returned Products
- Return Percentage

These metrics provide an overview of the return behavior within the dataset.

3. Statistical Analysis

Descriptive statistical analysis was performed to understand the characteristics of the dataset.

Measures Calculated

- Mean
- Median
- Standard Deviation
- Minimum and Maximum Values
- Quartiles

The analysis focused on:

- Product Price
- Customer Age
- Total Purchase Amount

Exploratory Data Analysis (EDA)

Several visualizations were created to identify patterns and trends:

Histogram

Used to analyze the distribution of product prices.

Bar Charts

Used to compare return rates across:

- Product Categories
- Payment Methods
- Age Groups

Box Plots

Used to identify relationships between:

- Product Price and Returns
- Quantity and Returns

Heatmap

Correlation analysis was performed to understand relationships between numerical variables. The statistical analysis helped identify patterns associated with product returns and customer behavior.

4. Return Pattern Investigation

The primary goal of this phase was to identify factors influencing product returns.

Product Category Analysis

Return rates were calculated for each product category. Categories with higher return percentages were identified as potential risk areas for the business.

Payment Method Analysis

Return rates were compared across different payment methods to determine whether customer payment preferences influence return behavior.

Age Group Analysis

Customers were segmented into age groups:

- 18–25
- 26–35
- 36–45
- 46–55
- 55+

Return rates were calculated for each group to identify customer segments with higher return tendencies.

Product Price Analysis

Product prices were compared between returned and non-returned orders. This helped determine whether expensive products are more likely to be returned.

5. Outlier Analysis

Outlier detection was performed using the Interquartile Range (IQR) method.

Objective

Identify unusual observations such as:

- Extremely expensive products
- Abnormally high purchase amounts
- Unusual customer purchasing behavior

Method Used

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

Records outside these boundaries were considered outliers.

Importance

Outlier analysis helps businesses detect unusual purchasing patterns that may influence return behavior and operational performance.

6. Hypothesis Testing

A Chi-Square Test of Independence was performed.

Hypotheses

Null Hypothesis (H_0):

Product Category and Returns are independent.

Alternative Hypothesis (H_1):

Product Category significantly influences Returns.

Test Used

Chi-Square Test

Decision Rule

- If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$
- If $p\text{-value} \geq 0.05 \rightarrow \text{Fail to Reject } H_0$

Interpretation

The statistical test helps determine whether product categories play a significant role in customer return behavior.

7. Customer Segmentation

Customer segmentation was performed to identify groups requiring business attention.

High-Return Customers

Customers with multiple returns were identified for further monitoring.

Premium Customers

Customers with high purchase amounts were categorized as premium customers.

Returning Customers

Customers who returned products were analyzed to understand potential issues with product quality or customer expectations.

Customer segmentation enables targeted interventions and personalized strategies for different customer groups.

8. Business Insights

Based on the analysis, several key insights were identified.

Which Factors Contribute Most to Returns?

- Product category significantly influences return behavior.
- Customer purchasing patterns impact return rates.
- Certain customer groups show higher return tendencies.

Which Categories Are Risky?

Product categories with the highest return rates should be prioritized for quality checks, supplier reviews, and product improvements.

Which Customer Groups Need Intervention?

- High-return customers
- Customers with repeated returns
- Premium customers requiring retention strategies

Which Payment Method Has the Highest Return Rate?

Analysis of payment methods helps identify customer segments associated with higher return behavior.

Which Age Group Returns Products the Most?

Age-group analysis highlights demographic segments that may require targeted communication and product recommendations.

9. Business Recommendations

Based on the findings, the following recommendations are proposed:

Improve Product Quality

- Strengthen quality control procedures.

- Conduct supplier performance evaluations.
- Reduce defects before products reach customers.

Enhance Product Information

- Provide accurate descriptions and specifications.
- Use high-quality images.
- Set realistic customer expectations.

Monitor High-Return Customers

- Identify customers with frequent returns.
- Investigate reasons behind repeated return behavior.

Focus on High-Risk Categories

- Review products with high return rates.
- Improve packaging and product presentation.

Optimize Inventory Management

- Track return-prone products.
- Adjust inventory planning accordingly.

Strengthen Customer Support

- Provide better pre-purchase guidance.
- Resolve customer concerns before returns occur.

10. Conclusion

This project successfully investigated product return behavior using statistical analysis, exploratory data analysis, hypothesis testing, and customer segmentation techniques.

The findings demonstrate that product category, customer characteristics, and purchasing behavior influence return rates. By implementing the recommended operational improvements, businesses can reduce return-related losses, improve customer satisfaction, and enhance overall profitability.

Future work may include predictive modeling and machine learning techniques to forecast returns and proactively reduce return risks.